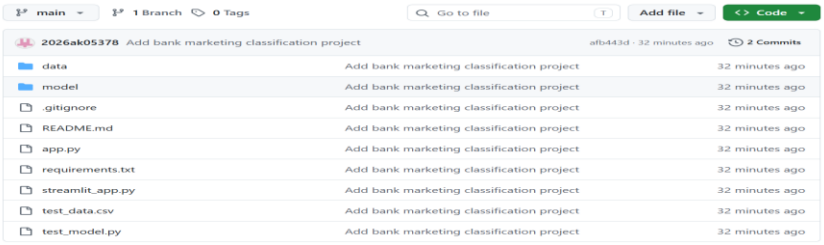
BITS Pilani – Work Integrated Learning Programmes Division

M.Tech (AIML) | Machine Learning (Assignment – 2)

Student: **Lexili J Alex** | BITS ID: **2026AK05378**

1. Mandatory Submission Links

Deliverable	Submission Details & Clickable URL
1. GitHub Repository Link	**URL:** https://github.com/2026ak05378/ML_Assignment_2/tree/main  *Contains: app.py, requirements.txt,data/ test_data.csv, README.md, model/ folder with saved .joblib classifiers, and train_models.py*
2. Live Streamlit App Link	**URL:** https://mlassignment2-6ybhavskvmucb2nusstsx5.streamlit.app/ Interactive web app deployed on Streamlit Community Cloud

2. Proof of Execution on BITS Virtual Lab

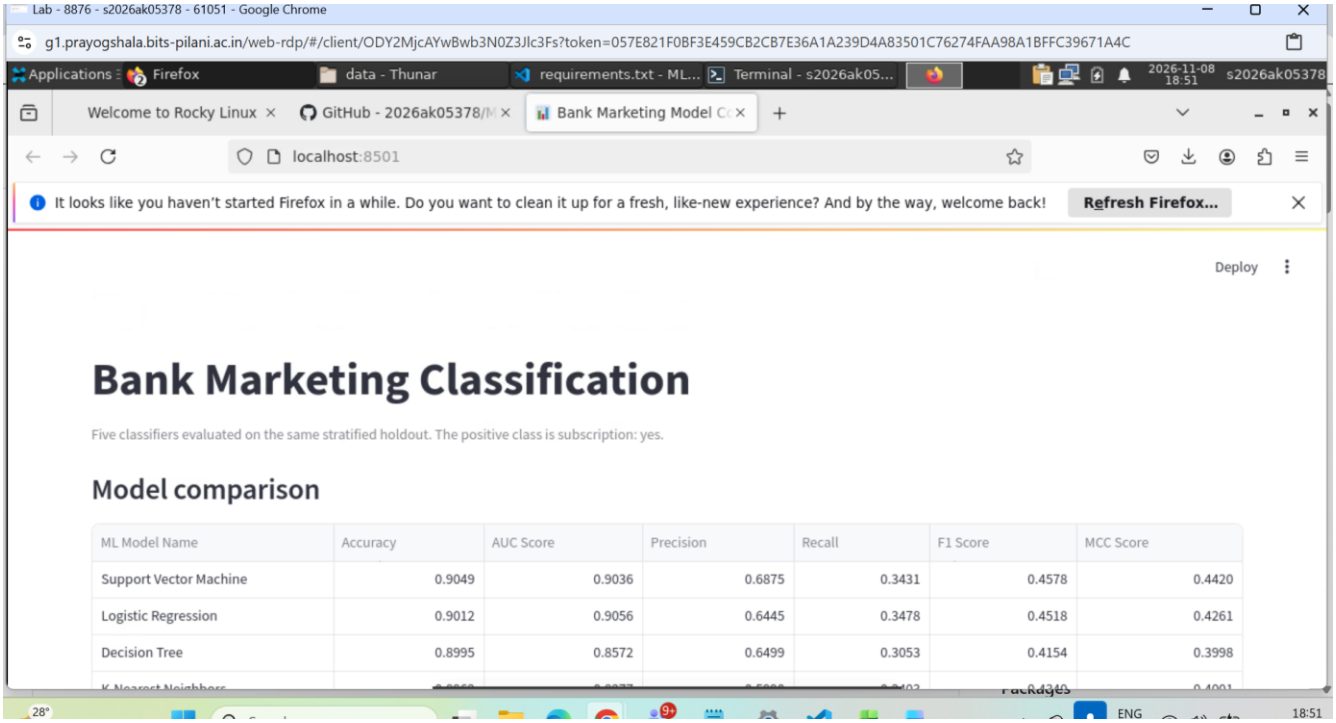


Figure 1: Verified execution of the deployed Streamlit application inside the BITS Prayogshala Virtual Lab environment.

3. GitHub README Content (Project Documentation)

a. Problem statement

This project predicts whether a bank customer will subscribe to a term deposit. The model uses customer details and information about the bank's contact with the customer. In the target column y, yes means the customer subscribed and no means they did not.

b. Dataset Description

The project uses the Bank Marketing bank-full.csv dataset, containing 45,211 records, 16 input features, and the binary target y. It has no missing values. The target is imbalanced: 39,922 clients did not subscribe and 5,289 did.
The 20% holdout in test_data.csv was created with stratification and random_state=42 (9,043 records). The remaining 80% is used for training. Categorical inputs are one-hot encoded and numeric inputs are standardized; these preprocessing steps are fitted on training data only.

Dataset Details

- **Dataset:** Bank Marketing Dataset (bank-full.csv) – UCI Machine Learning Repository.
- **Problem Type:** Binary Classification.
- **Target Variable:** y
  - yes – Customer subscribed to a term deposit.
  - no – Customer did not subscribe.
- **Data Preprocessing:** Handling categorical and numerical features using One-Hot Encoding and StandardScaler.
- **Exploratory Data Analysis (EDA):** Analysing customer characteristics, feature distributions, and target class imbalance.
- **Model Development:** Training and comparing different machine learning classification algorithms.
- **Model Evaluation:** Evaluating performance using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Matthews Correlation Coefficient (MCC).
- **Confusion Matrix:** Analysing correct and incorrect predictions for both classes.

c. GitHub Repository Link

- **Repository URL:** https://github.com/2026ak05378/ML_Assignment_2.git

(Contains app.py, requirements.txt, README.md, test_data.csv, and pre-trained models in model).

d. Models Used

All models use the same train/test split. Precision, recall, and F1 treat yes as the positive class. The SVM AUC is calculated from its decision scores.

Model configurations:

- Logistic Regression: max_iter=1000.
- Decision Tree: max_depth=5, random_state=42.
- K-Nearest Neighbors: n_neighbors=5.
- Gaussian Naive Bayes: Gaussian likelihood; encoded input is converted to a dense matrix for fitting.
- Support Vector Machine: RBF kernel, random_state=42.

The evaluator calculates accuracy, ROC AUC, precision, recall, F1, and Matthews correlation coefficient (MCC). AUC uses predicted probabilities where available and the SVM decision function otherwise.

ML Model Name	Accuracy	AUC	Precision	Recall	F1	MCC
Logistic Regression	0.9012	0.9056	0.6445	0.3478	0.4518	0.4261
Decision Tree	0.8995	0.8572	0.6499	0.3053	0.4154	0.3998
KNN	0.8962	0.8277	0.5990	0.3403	0.4340	0.4001
Naïve Bayes (Gaussian)	0.8548	0.8101	0.4059	0.5198	0.4559	0.3774
Support Vector Machine	0.9049	0.9036	0.6875	0.3431	0.4578	0.4420

Observations

ML Model Name	Observation about model performance
Logistic Regression	Achieves the highest **AUC (0.9056),** indicating the strongest ranking performance, with accuracy close to the SVM. Recall is modest at **0.3478**.
Decision Tree	Has **0.8995** accuracy, but lower **AUC (0.8572)** and the lowest recall **(0.3053),** so it misses more actual subscribers at the default threshold.
KNN	Reaches **0.8962** accuracy, but its **AUC (0.8277)** is the lowest among the models. Precision and recall are both moderate.
Naive Bayes	Has the highest recall **(0.5198)** and F1 **(0.4559)**, but lower accuracy and precision. It catches more subscribers at the cost of more false positives.
Support Vector Machine	Leads in accuracy **(0.9049),** precision **(0.6875),** and MCC **(0.4420)**, with AUC close to Logistic Regression. Its recall **(0.3431)** remains modest.

Overall winner:

Support Vector Machine by accuracy, precision, and MCC. There is no single winner across every metric: Logistic Regression has the best AUC, while Gaussian Naive Bayes has the best recall and F1. Given the class imbalance, accuracy alone should not determine model choice.